

Quadratic equations



1 $x = -12$

2

a $x = \pm\sqrt{2}$

b $x = \pm 5$

c $x = \pm 11$

d $x = \pm 7\sqrt{6}$

e $x = \pm 11$

f $x = \pm 5$

g $x = \pm 8$

h $x = \pm 15$

i $x = 4, -10$

j $x = 11, -5$

k $x = 6 \pm \sqrt{2}$

l $x = -7, 11$

m $x = 16, -2$

n $x = -2, 16$

o $x = \frac{7}{8}, \frac{25}{8}$

p $x = \pm \frac{4}{9}$

3

a $x = 0, -7$

b $x = 0, \frac{9}{2}$

c $x = \frac{9}{10}$

d $x = \frac{9}{4}$

e $x = \frac{3}{7}$

f $x = 4, 2$

g $x = 6, -7$

h $x = \frac{5}{8}, \frac{7}{3}$

i $x = -\frac{8}{3}, \frac{7}{5}$

j $x = \frac{17}{3}, \frac{d}{2}$

4

a $y = \pm 5$

b $y = \pm \frac{6}{5}$

c $k = \pm 2$

d $k = \pm \frac{4}{9}$

e $v = \pm \frac{8}{5}$

f $p = \pm 10$

g $m = 0, -5$

h $m = 0, -5$

5

$x = \pm \sqrt{\frac{c}{a}}$

Applications

6 $k = -16$

7 $a = 13$

8 $t = 14, t = 10$

9

a $x = 0, 13$

b 13 cm